EXPLORING BIOMASS QUALITY WITHIN THE BIOMASS-TO-BIOFUEL SUPPLY CHAIN USING PRINCIPAL COMPONENT ANALYSIS

By

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MASTER OF SCIENCE IN ADVANCED MANUFACTURING AND ENTERPRISE PROCESS ENGINEERING

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AGENDA

- Introduction
- Literature Review
- Methodology
- Case Study
- Conclusions
- Future Areas of Research

INTRODUCTION

What is biomass? Why is biomass quality so important?

INTRODUCTION Biomass

- Biomass is, by definition, organic matter, especially plant matter, and it can be converted to fuel.

- *First generation*: edible crops such as sorghum and corn
  Second generation: non-edible crops such as wood chips, switchgrass, Miscanthus, organic waste, and food crop waste (e.g., corn stover)

SWITCHGRASS

CORN STOVER

MISCANTHUS

SORGHUM

Third generation: algae, a low cost, high energy, and entirely renewable source of energy

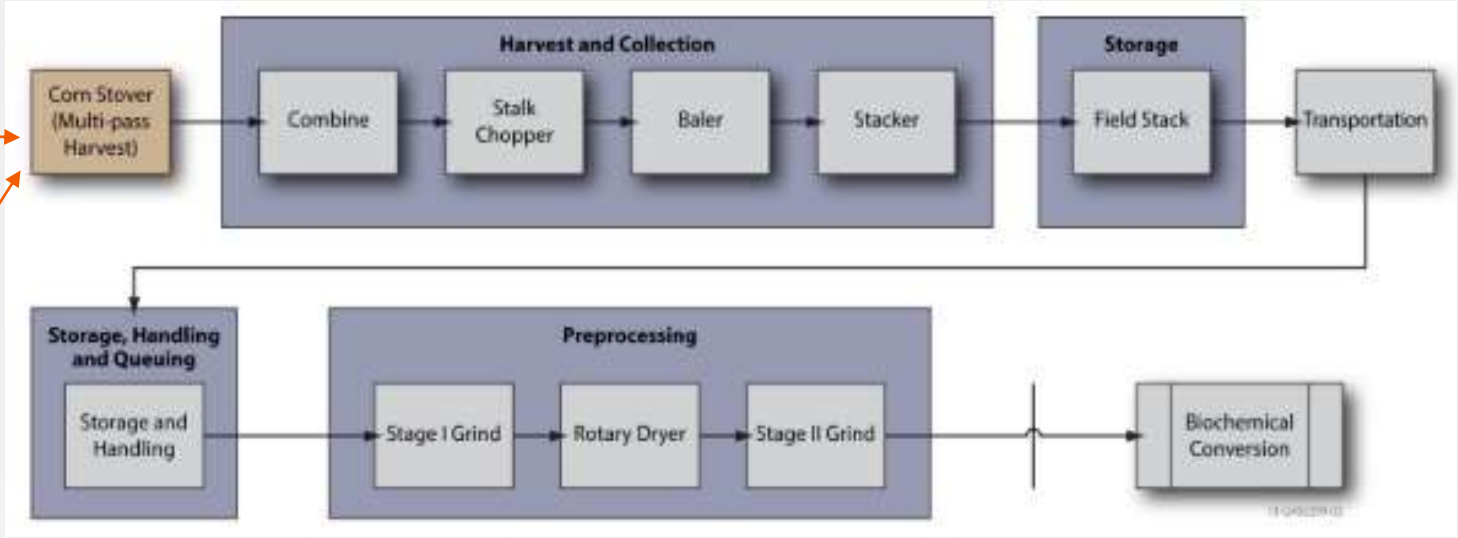
- Biomass is the largest single source of renewable energy and, in 2016 was found to have the potential to replace 30% of the then-current petroleum demand (Goss, Thesis, and Rials 2016)

INTRODUCTION Biomass Physical and Chemical Properties

Physical	Moisture content	Storage durability Dry-matter losses Low NCV Self ignition
	Bulk density	Fuel logistics (storage, transport, handling) costs
	Ash content	Dust, particle emissions Ash utilization/disposal costs
	Particle dimension and size distribution	Determines fuel feeding system Determines combustion technology Drying properties Dust formation Operational safety during fuel conveying
Chemical	Carbon C	GCV (positive)
	Hydrogen H	GCV (positive)
	Oxygen O	GCV (negative)
	Chlorine Cl	Corrosion
	Nitrogen N	NOx, N2O, HCN emissions
	Sulfur S	SOx emissions, corrosion
	Fluor F	HF emissions Corrosion
	Potassium, K	Corrosion (heat exchangers, superheaters) Lowering of ash melting temperatures Aerosol formation Ash utilization (plant nutrient)
	Sodium, Na	Corrosion (heat exchangers, superheaters) Lowering ash melting temperatures Aerosol formation
	Magnesium, Mg	Increase of ash melting temperature Ash utilization (plant nutrient)
	Calcium, Ca	Increase of ash melting temperature Ash utilization (plant nutrient)
	Phosphorus, P	Increase in ash meting point Ash utilization (plant nutrient)
	Heavy metals	Emissions of pollutants Ash utilization and disposal issues Aerosol formation

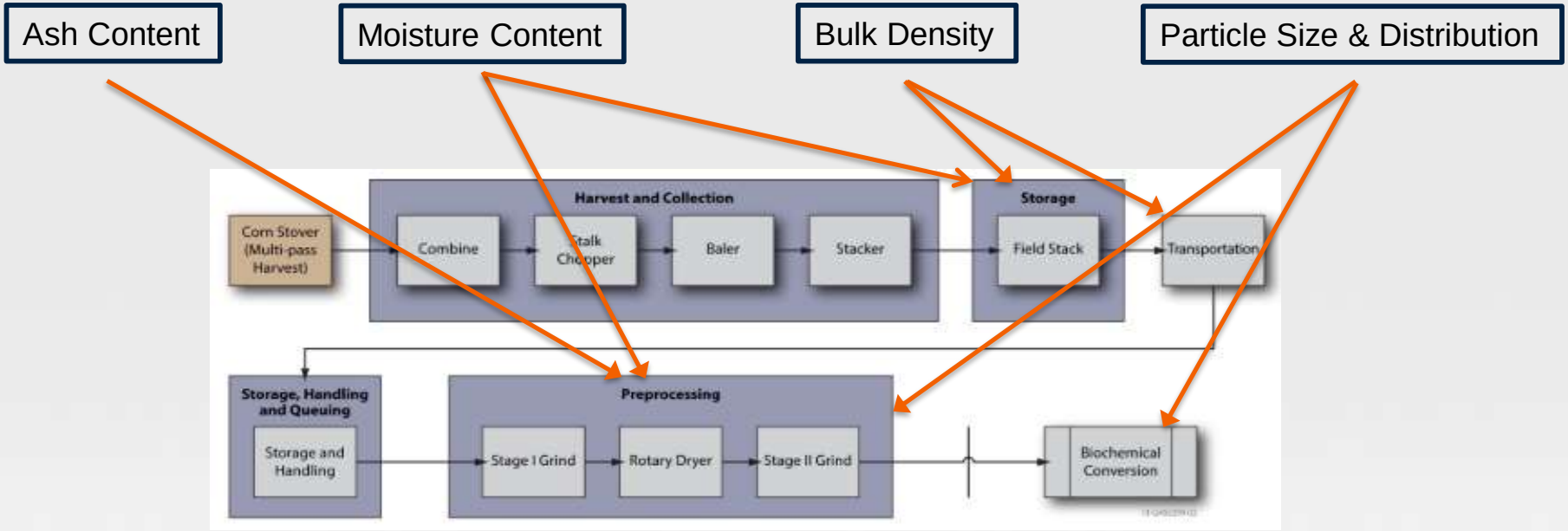
Source: European Biomass Association et al. (2000) and Loo van and Koppejan (2004)
Note: GCV: Gross calorific value, NCV: Net calorific value

INTRODUCTION Processes within the Biomass-to-Biofuel Supply Chain



- It is important to understand sources of variation and correlations between variables to optimize biomass-to-biofuel supply chains

INTRODUCTION Considering Biomass Quality in Supply Chain Optimization



- Uncertainty in biomass quality is often present along the supply chain and can ultimately lead to unexpected variability when analyzing profitability of bioenergy investments.

LITERATURE REVIEW

What is the current state of biomass quality research?

LITERATURE REVIEW

Current Research Considering Biomass Quality

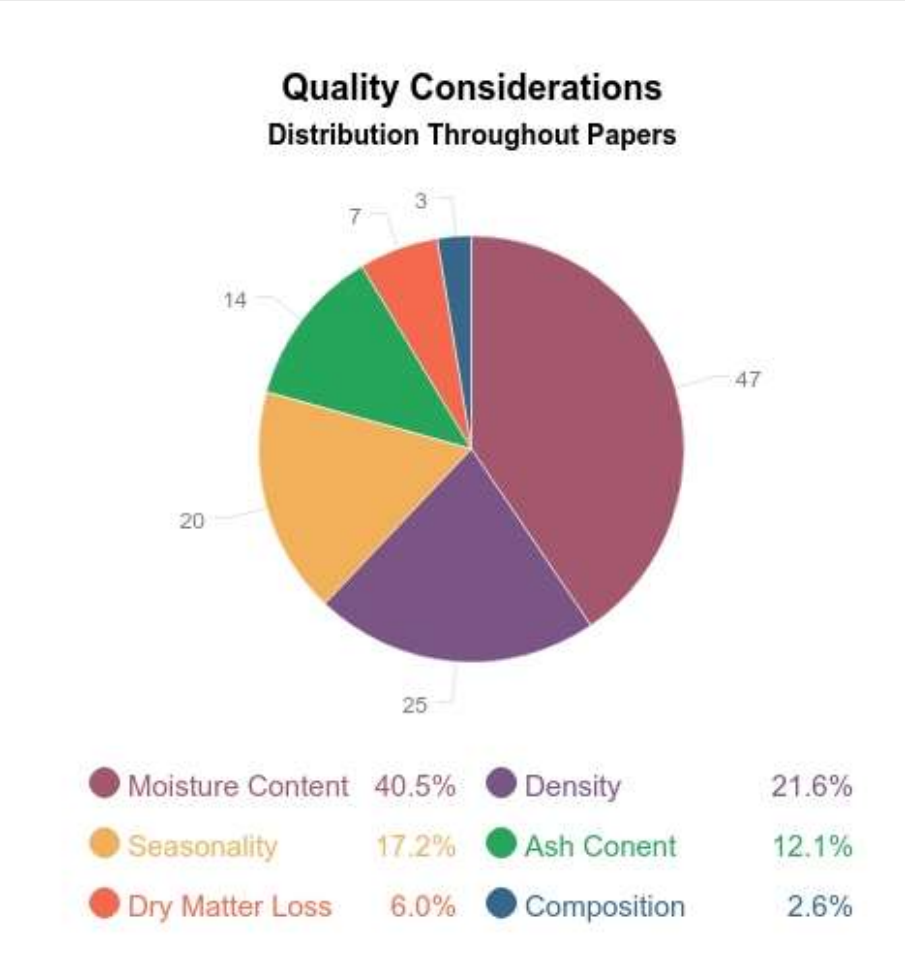
Year, Author/ Optimization Method		Mathematical Modeling								Quality Characteristic Considered
		Deterministic		Stochastic		Heuristic		Non-Linear		
		Single Objective	Multi-Objective	Single Objective	Multi-Objective	Single Objective	Multi-Objective	Single Objective	Multi-Objective	
2017	Belart	X								Moisture Content
2016	Castillo	X								Ash, Moisture Content
2014	Hartley	X								Density, Seasonality
2010	Huang	X								Moisture Content
2014	Lin	X								Moisture Content, Density
2012	Marvin	X								Moisture Content
2014	Sosa	X								Moisture Content, Density
2018	Hernandez	X								Moisture Conent, Ash Content, Seasonality
2017	Giuliano	X								Lignin Conent, Seasonality
2012	Acuna	X								Moisture Content
2014	Khorshidi	X								Moisture Conent
2012	Alam	X								Seasonality
2018	Malladi	X								Seasonality
2015	Taskhiri	X								Moisture Conent
2011	Van Dyken	X								Moisture Content, Density
2009	Alfonso		X							Moisture Content, Ash Content, Seasonality
2017	Arora		X							Moisture Content, Ash Content
2015	Cambero		X							Moisture Content, Ash Content, Density
2016	Cambero		X							Moisture Content, Ash Content, Density
2012	Cucek		X							Moisture Content, Density
2013	Kanzian		X							Dry Matter Loss
2014	Perez-Fortes		X							Moisture Content, Ash Content, Density, Seasonality, Dry Matter Losses
2017	Tan		X							Moisture Conent, Seasonality
2012	You		X							Moisture Content, Ash Content, Density, Seasonality
2016	Azadeh			X						Moisture Conent
2014	Azadeh			X						Moisture Conent
2017	Castillo			X						Moisture Content, Ash Content, Seasonality
2012	Chen			X						Moisture Conent
2014	Gonzalez-Salazar			X						Moisture Content, Density
2016	Lopez-Diaz			X						Density
2014	Li			X						Moisture Content
2015	Shabani			X						Moisture Content, Ash Content
2015	Zamar			X						Moisture Content
2012	Chen				X					Moisture Conent
2017	Cobuloglu				X					Seasonality
2012	Gebreslassie				X					Moisture Conent, Density, Seasonality,
2002	Gigler				X					Moisture Conent, Particle Size
2014	Marufuzzman				X					Energy Loss
2018	Xie				X					Moisture Content, Density
2007	Gronalt					X				Moisture Content, Seasonality
2008	Reche-Lopez					X				Density
2015	Poudel					X				Density
2016	Poudel					X				Seasonality
2017	Quddus					X				Moisture Content, Density, Seasonality
2013	Gomez-Gonzalez					X				Density
2010	Vera					X				Density
2017	Yang					X				Composition, Quality Control
2012	You						X			Moisture Content, Ash Content, Density, Seasonality
2016	Hernandez-Calderon							X		Density
2012	Shabani							X		Moisture Content, Ash Content, Density
2017	Tan								X	Moisture Content, Density

Year, Author/ Optimization Method		Simulation						Quality Characteristic Considered
		Discrete Event Simulation		Continuous Simulation	Agent-based Simulation	Simulation-based Optimization Approach		
		Single Objective	Multi-Objective			Single Objective	Multi-Objective	
2013	Amundson	X						Moisture Content, Density
2017	Kishita	X						Moisture Content
2016	Pinho	X						Moisture Content
2011	Mobini	X						Moisture Conent
2015	Windisch	X						Moisture Content
2012	Woli		X					Moisture Content, Seasonality
2015	Bai			X				Density
2014	Mobini			X				Moisture Content, Density
2017	Sándor			X				Moisture Content, Dry Matter Loss
2017	Shu				X			Density
2014	Singh				X			Seasonality
2013	Ebadian					X		Moisture Content, Seasonality, Dry Matter Loss
2011	Faulkner					X		Density, Seasonality
2013	Sharma					X		Moisture Content, Seasonality
2017	Chavez					X		Moisture Content, Ash Content, Dry Matter Loss

- Eighty-eight scientific publications identified as significant references in terms of modeling and optimization have been reviewed. The publications range from 2002-2018.

LITERATURE REVIEW

Current Research Considering Biomass Quality



- A recent study in Germany conducted extensive field trials, which indicate that five months of storage results in dry matter loss of 11.1% and energy losses of up to 11.3% [50].

LITERATURE REVIEW

Modeling Approaches

Mathematical Modeling							
Deterministic		Stochastic		Heuristic		Non-Linear	
Single Objective	Multi-Objective	Single Objective	Multi-Objective	Single Objective	Multi-Objective	Single Objective	Multi-Objective

Simulation					
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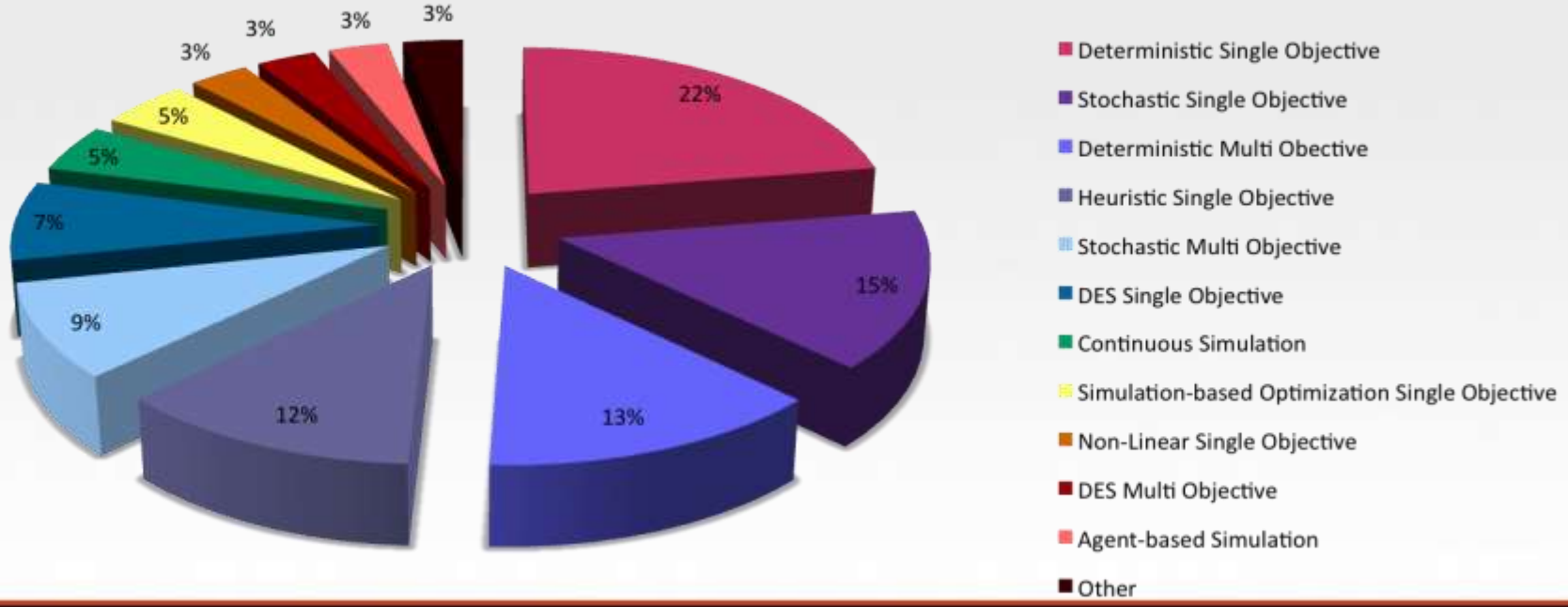


LITERATURE REVIEW

Challenges and Gaps in the Literature

- Cost reductions along supply chain operations ranging from 6%-31% were found through several studies quantifying biomass quality effects [21], [36], [38], [47]. Further, it is important to account for the uncertainties and variability inherent to biomass quality during both design and management phases of the supply chain [22].

- A deeper exploration into the quantification and control of the impacts of ash, biomass composition, and quality variability on the biomass supply chain is needed; these specific characteristics were not addressed in the literature as much as moisture content, bulk density, and seasonality were.

PROBLEM & THESIS STATEMENT

- Problem Statement

Ignoring the uncertainty and variability of biomass quality could lead to inaccurate results and therefore discrepancies in supply chain logistics, which can be a costly mistake. Biomass quality control methods and standardization protocol are underdeveloped in current research and must be taken in consideration when analyzing profitability of bioenergy investments.

- Thesis Statement

The goal of this thesis is to investigate how biomass physical and chemical composition effect supply chain logistics. This thesis aims to provide both theoretical and practical insights on the impact of biomass variability on the performance of the preprocessing systems through the use of statistical techniques applied to a realistic case study.

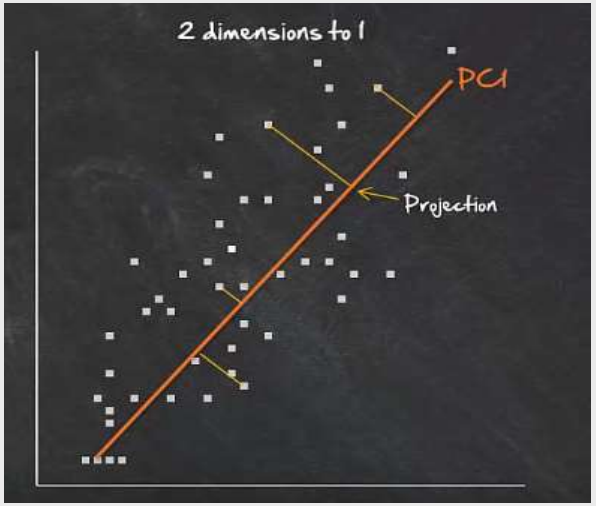
METHODOLOGY

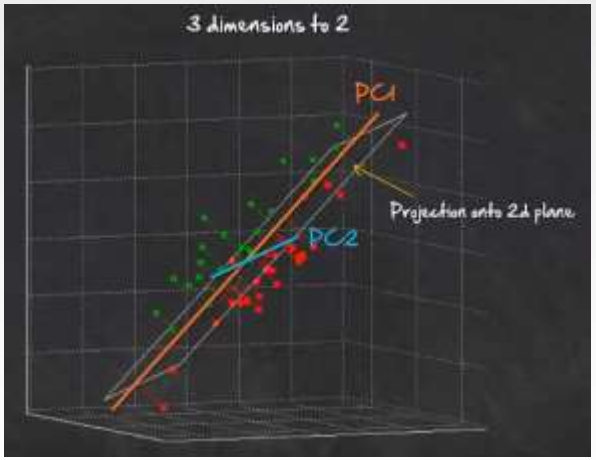
What technique can I use to contribute to current biomass quality research?

METHODOLOGY Principal Component Analysis

- Procedure to measure the variance in a large set of data using a smaller number of uncorrelated components, called "principal components."

- This method of analysis is useful to uncover patterns and can reduce a large amount of data into a subset of significant relationships.





METHODOLOGY Principal Component Analysis

- PCA essentially works in steps toward solving an eigenvalue/eigenvector problem to reduce the dimensions of a dataset.

1. First, calculate the first principal component, a linear combination of $x_1, x_2, \ldots, x_p$ (i.e., $y_1 = a_{11}x_1 + a_{12}x_2 + \ldots + a_{1p}x_p$) such that the variance of y_1 is maximized

2. In a similar fashion, moving to the next set of variables, y_2 will be calculated and account for the second largest percentage of variance between all components.

3. Further, the PCA will find the optimal weight vector for the first component ($a_{11}, a_{12}, \ldots, a_{1p}$) and its' associated variance; this calculation will be performed for each principal component found and will be maximized, subject to the constraint that it must be uncorrelated with the previous components.

4. From these variable weight vectors and associated variances for each principal component, an indication of how each variable contributes to the variance is seen.

METHODOLOGY Expanding on Previous Works

- In research efforts specifically aimed toward expanding knowledge of biomass preprocessing variables, this tool has proven to be advantageous.

- Aboytes et al. investigates direct impact of storage days on biomass chemical composition

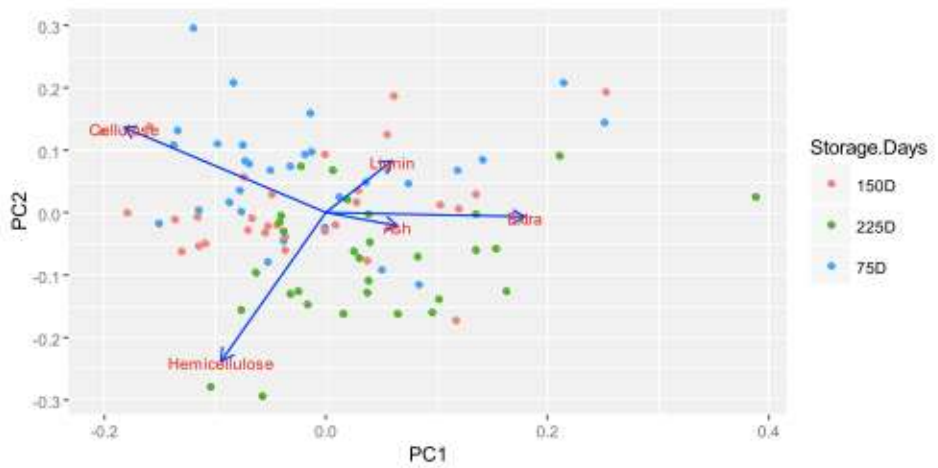


Figure 6. Bi-plot chart for analysis of storage days.

- Shukla et al. [103] used similar techniques to assess the relationship between soil attributes and biomass yield. Knowing biomass and grain yields depend on complex interactions among spatially variable physical and chemical properties of soil, PCA was used to determine the ideal soil treatment for maximized yield.

METHODOLOGY Expanding on Previous Works

- This study uses multivariate analysis to summarize the relationship between biomass properties and controllable factors along the biomass-to-biofuel supply chain.

- This paper differentiates from previous works through exploration of biomass type and moisture content effects on preprocessing logistics. These factors have not been studied in published works using the PCA technique and will provide valuable insight towards optimizing the preprocessing of biomass and lowering logistics costs and energy consumption.

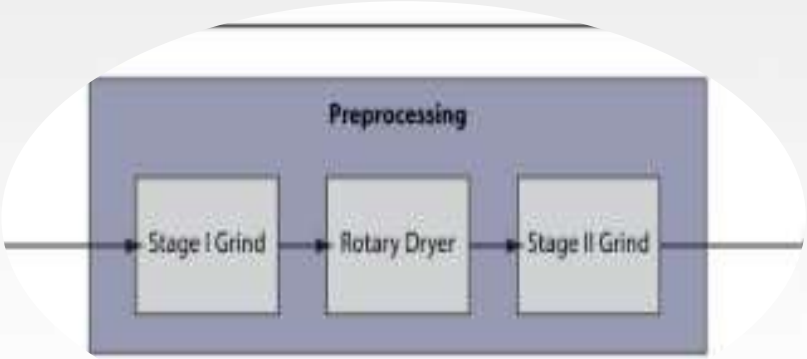
CASE STUDY

Applying methodology to collected data

CASE STUDY Processing Demonstration Unit at INL

- A unique capability of the Feedstock PDU is to test and develop fully integrated and instrumented industrial-scale preprocessing systems. Continuous processing and data collections enables:

- Identification of preprocessing bottlenecks
- Exploration of the interaction and synergies of preprocessing operations affecting feedstock specification and conversion performance
- Optimization of integrated preprocessing system



CASE STUDY Data Collection

- Data was obtained from INL report titled "Enabling Sustainable Landscape Design for Continual Improvement of Operating Bioenergy Supply Systems" from June 30, 2018

Table 1: Complete Data for Analysis

Measured Throughput	Measured Energy Consumption	Logistics Cost($/dry ton)	Preprocessing Cost($/dry ton)	Biomass	Incoming Moisture
4.54	19.91	93	14.66	Switchgrass	15
5.24	6.44	45.73	11.24	Switchgrass	10
4.97	6.58	49.09	11.88	Switchgrass	13.5
4.54	19.91	48.65	13.47	Switchgrass	14.5
4.97	6.58	49.36	11.91	Switchgrass	13.7
4.97	6.58	43.45	11.66	Switchgrass	12.5
4.54	19.91	70.89	14.37	Switchgrass	16.5
4.97	6.58	57.02	11.74	Switchgrass	12
3.89	7.09	93	14.66	Switchgrass	13
5.03	6.14	45.73	11.24	Switchgrass	8
4.67	6.73	49.09	11.88	Switchgrass	11.5
4.67	6.73	48.65	13.47	Switchgrass	12.5
4.67	6.73	49.36	11.91	Switchgrass	11.7
4.86	6.38	43.45	11.66	Switchgrass	10.5
3.89	7.09	70.89	14.37	Switchgrass	14.5
4.86	6.38	57.02	11.74	Switchgrass	10
3.82	13.29	49.03	15.42	Corn Stover	14.17
4.28	15.83	39.55	13.74	Corn Stover	10.5
3.82	13.29	41.23	15.25	Corn Stover	13.64
4.04	14.33	40.25	14.37	Corn Stover	11.5
3.95	15.05	51.75	15.36	Corn Stover	12.5
3.95	15.05	49.48	15.28	Corn Stover	12.5
3.69	13.15	59.71	16.26	Corn Stover	17.17
3.45	15.49	55.4	16.82	Corn Stover	20.5
3.69	13.15	58.82	16.12	Corn Stover	17.17
4.06	18.09	49.03	15.42	Corn Stover	12.17
4.91	13.39	39.55	13.74	Corn Stover	8.5
4.06	18.09	41.23	15.25	Corn Stover	11.64
4.4	13.85	40.25	14.37	Corn Stover	9.5
4.16	18.4	51.75	15.36	Corn Stover	10.5
4.16	18.4	49.48	15.28	Corn Stover	10.5
3.64	17.4	59.71	16.26	Corn Stover	15.17
3.57	15.43	55.4	16.82	Corn Stover	18.5
3.64	17.4	58.82	16.12	Corn Stover	15.17

CASE STUDY Implementation of the PCA

Table 2: Factors and Variables

Factors	No. Groups	Variables
Biomass	2	throughput, energy consumption, logistics cost, preprocessing cost
Moisture Percentage	varies	throughput, energy consumption, logistics cost, preprocessing cost

CASE STUDY Implementation of the PCA

STEP 1: EIGENANALYSIS OF CORRELATION MATRIX

Table 3. Correlation Coefficient Matrix

Pearson's Correlation Value	Measured Throughput	Measured Energy Consumption	Logistics Cost	Preprocessing Cost
Measured Throughput	-	-0.557	-0.232	-0.934
Measured Energy Consumption	-0.557	-	0.093	0.739
Logistics Cost	-0.232	0.093	-	0.24
Preprocessing Cost	-0.934	0.739	0.24	-

Table 4. Eigenvalues of Correlation Matrix

Eigenanalysis of the Correlation Matrix

Eigenvalue	2.5682	0.9471	0.4508	0.0339
Proportion	0.642	0.237	0.113	0.008
Cumulative	0.642	0.879	0.992	1.000

CASE STUDY Implementation of the PCA

STEP 2: OBTAIN EIGENVECTORS AND PRINCIPAL COMPONENT VALUES

Table 5. Eigenvectors

Eigenvectors

Variable	PC1	PC2	PC3	PC4
Measured Throughput	-0.574	-0.031	0.561	0.596
Measured Energy Consumption	0.504	0.267	0.785	-0.241
Logistics Cost($/dry ton)	0.208	-0.960	0.185	-0.024
Preprocessing Cost($/dry ton)	0.611	0.078	-0.184	0.766

The more correlated variables within each principal component were:

PC1- Preprocessing Cost, Measured Throughput, and Measured Energy Consumption

PC2- Logistics Cost and Measured Energy Consumption

PC3- Measured Energy Consumption and Measured Throughput

PC4- Preprocessing Cost and Measured Throughput

CASE STUDY Implementation of the PCA

STEP 3: OBTAIN SCORE AND SCREE PLOTS TO ANALYS VARIANCE ACCOUNTED FOR BY PRINCIPAL COMPONENTS

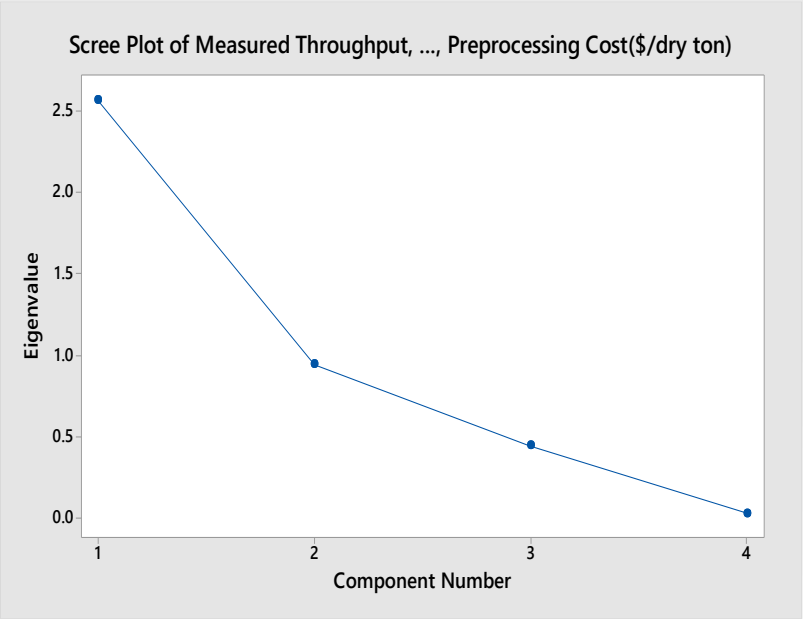


Figure 8. Scree Plot of PCA Variables

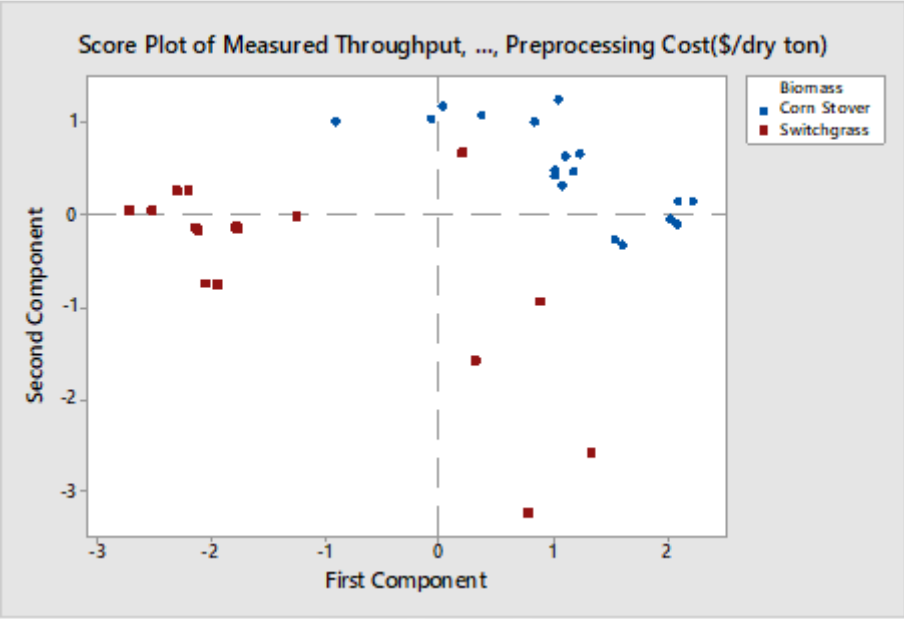


Figure 9. Score Plot of Variables

CASE STUDY

Using PCA to Understand Biomass Preprocessing Properties

STEP 4: GENERATE MONOPLOT TO ANALYZE VARIABLE RELATIONSHIPS

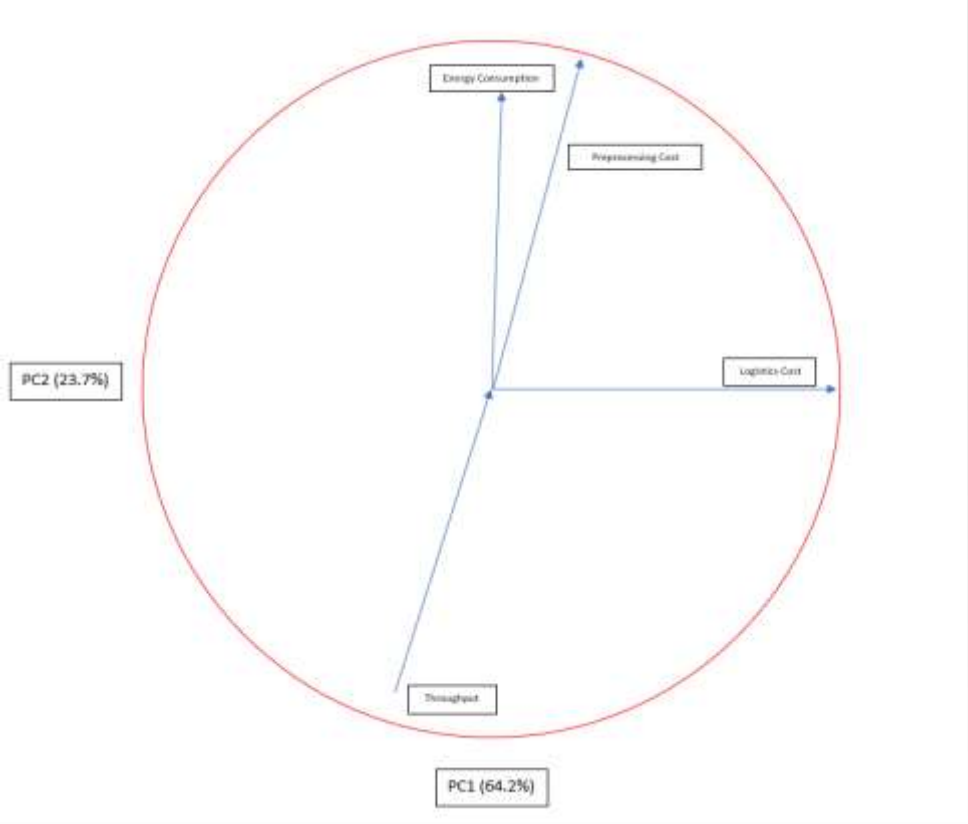


Figure 10. Monoplot of Variables

- The angles between the vectors approximates the correlation between the variables, with a small angle indicative of a positive correlation, no correlation would show a perpendicular vector, and an angle closer to 180 degrees is interpreted as a negative correlation.
- The length of each vector shows how well each variable is represented in the plot, with the poorly represented vectors being shorter.

CASE STUDY

Using PCA to Understand Biomass Preprocessing Properties

STEP 5: GENERATE BIPLOT TO ANALYZE VARIABLE RELATIONSHIPS WITH RESPECT TO FACTORS

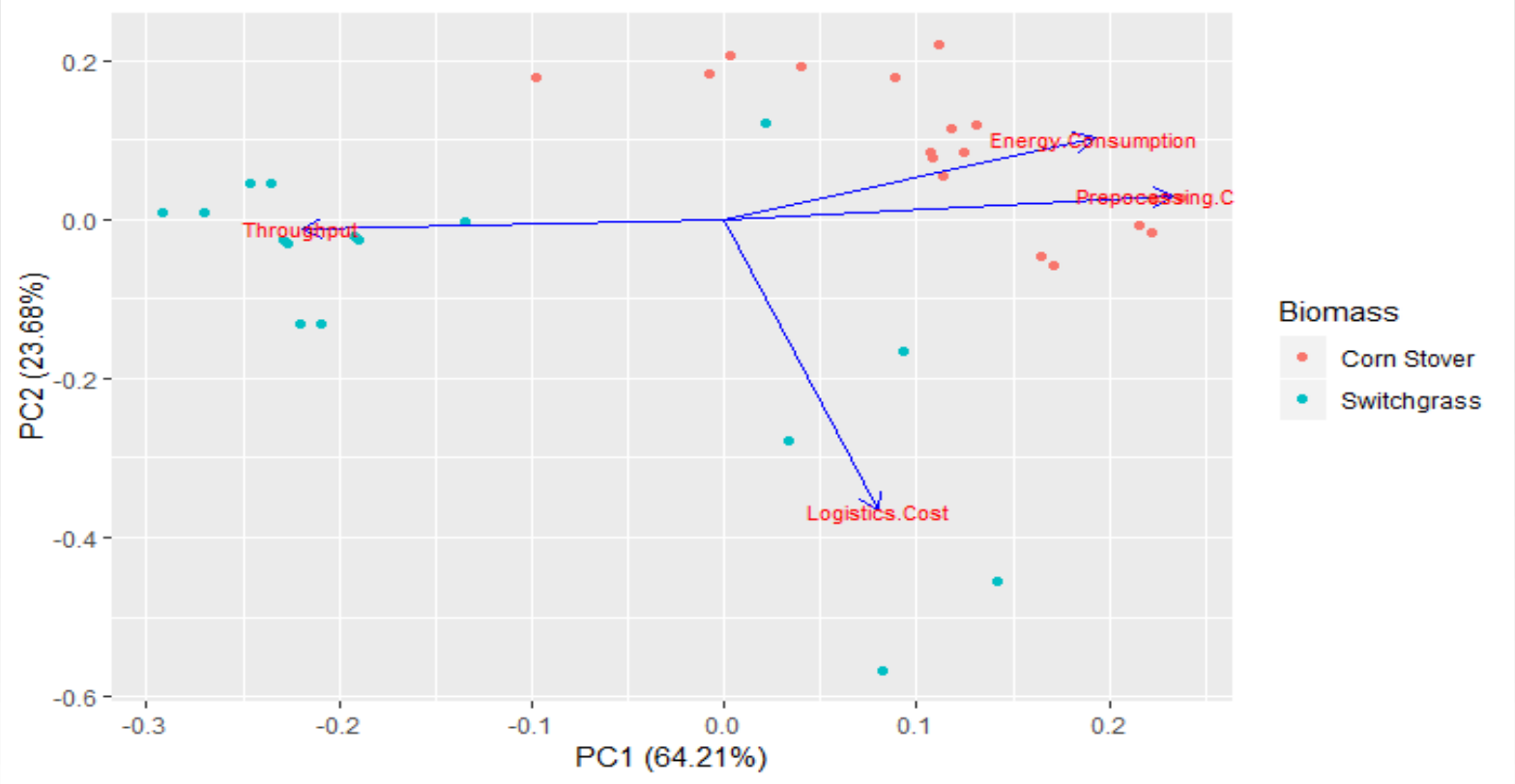


Figure 11. Biplot of Variables against Biomass Type

CASE STUDY

Using PCA to Understand Biomass Preprocessing Properties

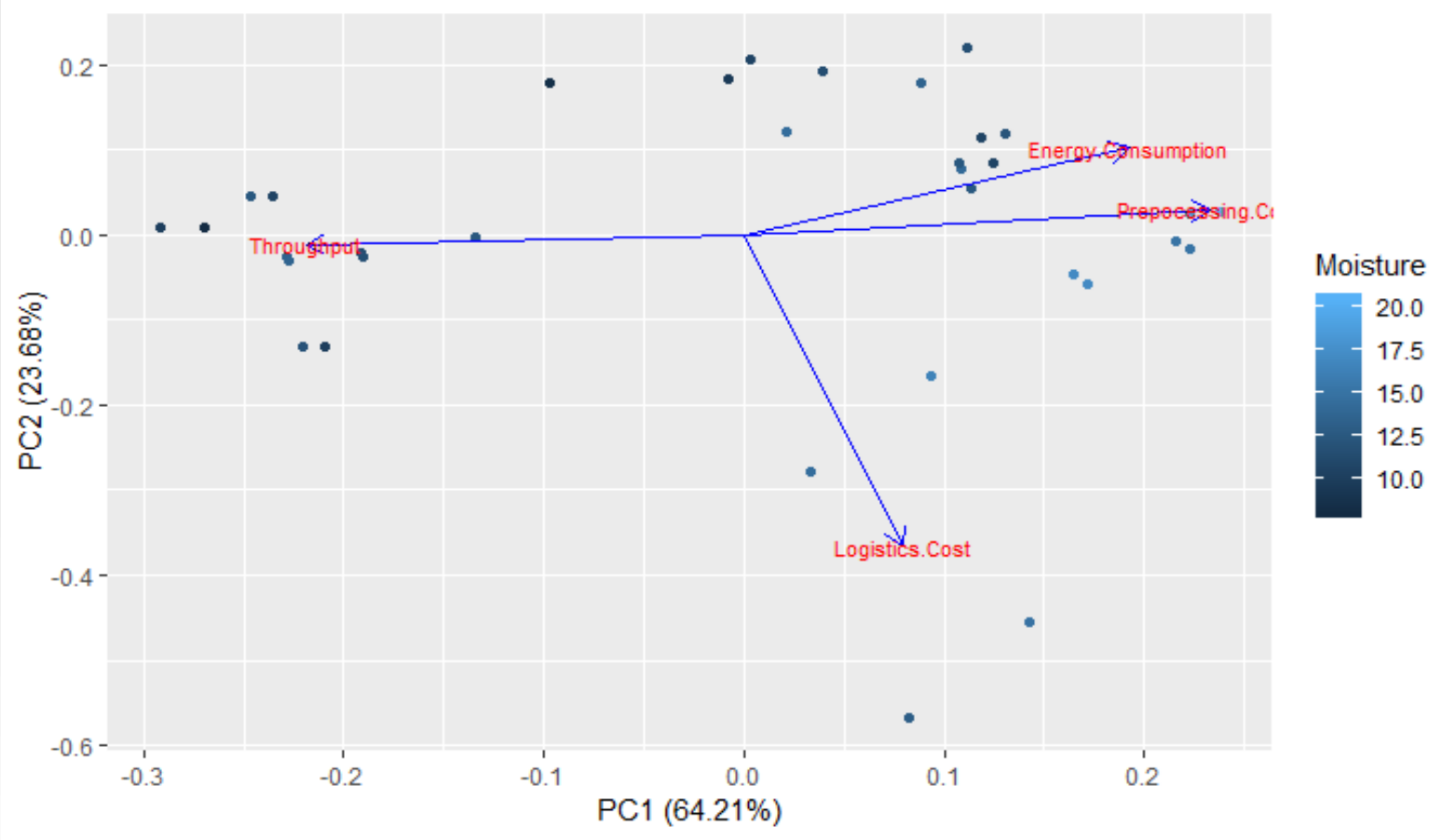
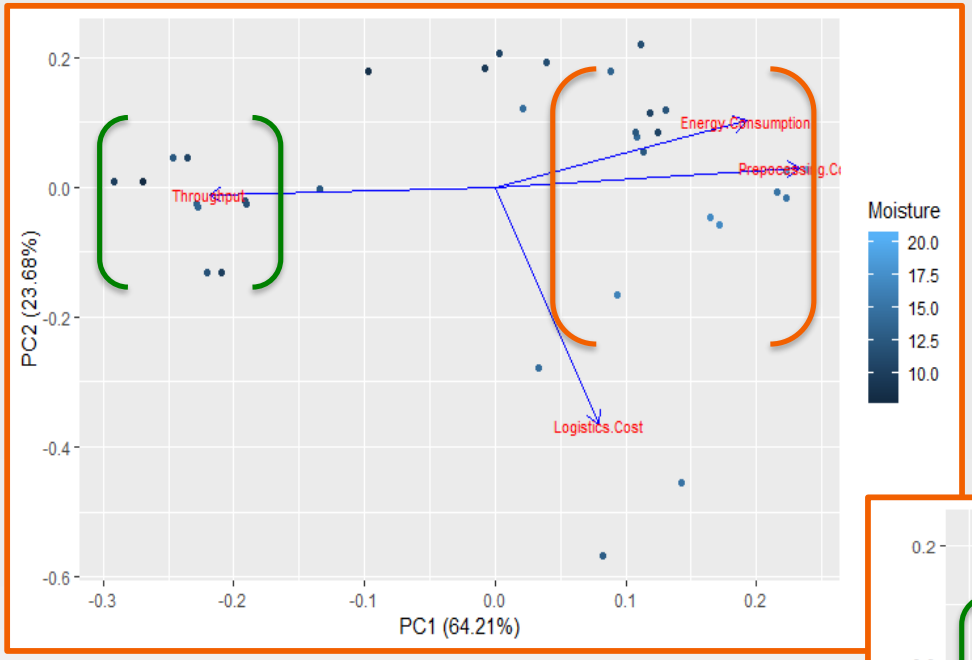
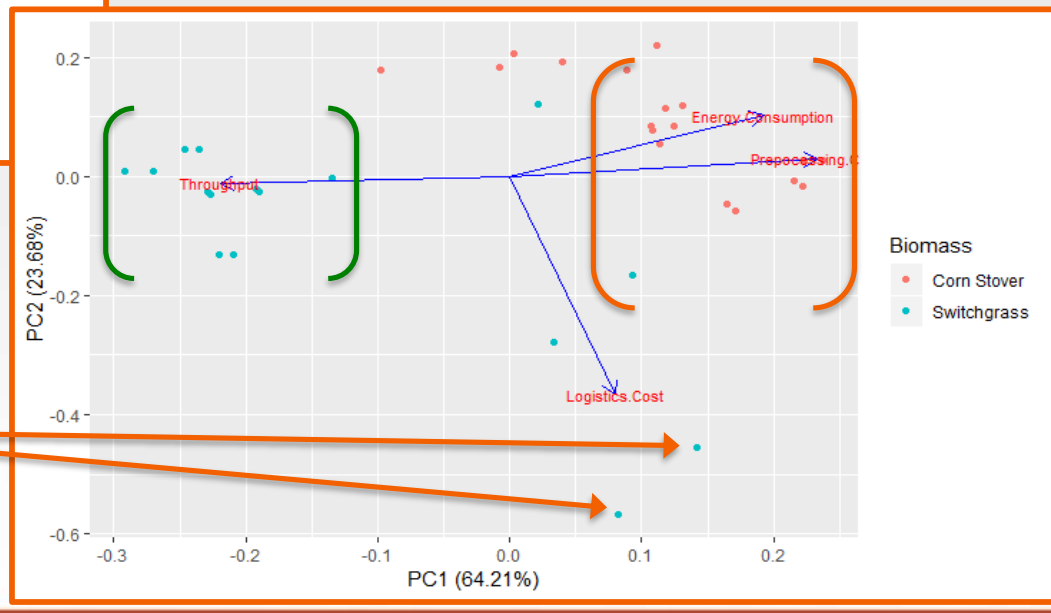


Figure 12. Biplot of Variables against Moisture Contents

CASE STUDY Significant Findings



- Switchgrass with lower moisture content between 0-15 percent, will return the optimal throughput and lower logistics and preprocessing costs.



- However, several outliers appear to stray from the clusters

CONCLUSIONS

What have I learned through this research?

CASE STUDY Significant Findings

- All outlier points are found to have originated from Field “Merk 4” and Field “Merk 13”
- Variability in logistics cost are due to cumulative impact of variability in harvesting methods, field efficiency, field shape, yield, biomass quality and bulk density of material.
- These fields returned the highest logistics costs due mainly to a larger grower payment and harvest location; the discrepancy is not due to our main factors and further confirms findings.

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3.69	13.15	59.71	16.26	Corn Stover	17.17
3.45	15.49	55.4	16.82	Corn Stover	20.5
3.69	13.15	58.82	16.12	Corn Stover	17.17
4.06	18.09	49.03	15.42	Corn Stover	12.17
4.91	13.39	39.55	13.74	Corn Stover	8.5
4.06	18.09	41.23	15.25	Corn Stover	11.64
4.4	13.85	40.25	14.37	Corn Stover	9.5
4.16	18.4	51.75	15.36	Corn Stover	10.5
4.16	18.4	49.48	15.28	Corn Stover	10.5
3.64	17.4	59.71	16.26	Corn Stover	15.17
3.57	15.43	55.4	16.82	Corn Stover	18.5
3.64	17.4	58.82	16.12	Corn Stover	15.17

CONCLUSIONS

- A total of 88 published articles written in English are reviewed and classified based on modeling approach and quality characteristic(s) are considered. It is concluded that the majority of papers discussed utilized a deterministic, single objective approach, while mainly accounting for the moisture content of biomass, as opposed to other biomass quality attributes.

- The case study used to further investigate the findings within the current research survey is conducted using Principal Component Analysis (PCA) through Minitab and R software. The graphical results indicate that decreasing costs and energy, while increasing throughput, may be achieved when switchgrass is used rather than corn stover. The PCA results for moisture content show that lower moisture content between 0-15% increase throughput and decrease logistics cost, energy consumption, and preprocessing cost

- Biomass quality has had a limited focus in past research, but holds an important role in operations such as harvest and collection, storage, preprocessing and transportation.

FUTURE AREAS OF RESEARCH

How can researchers expand upon these findings?

FUTURE AREAS OF RESEARCH

- A topic of discussion worth noting is the standardization of biomass quality and how that may affect the logistics of supply chain operations. Further research on the impact of quality control policies and standardization is essential to establish the necessity of such implementations.

- There is opportunity for research interests on exploring multistage stochastic programming methods, as well as simulation-based optimization approaches on biofuel supply chain optimization; these approaches allow for the modeling of complexities associated with the biomass supply chain and variability in biomass quality.

- Performing a PCA on additional data taken from the PDU is highly recommended; factors such as screen size, ash content, feed rate, etc. should be considered to further summarize ideal setup to maximize throughput and reduce cost and energy consumption.

FUTURE AREAS OF RESEARCH

- PCA becomes more useful as the number of variables increase; including more variables such as bulk density, ash content, particle size, and many others, will return principal components which can summarize a wider range of implications.

- The research and conclusions presented within this thesis are meant to further the research and development of biomass-to- biofuel supply chains and ultimately aim towards a society that relies on sustainable and renewable sources of energy.

THANK YOU FOR YOUR TIME

Questions?

LITERATURE REVIEW

Current Research Considering Biomass Quality

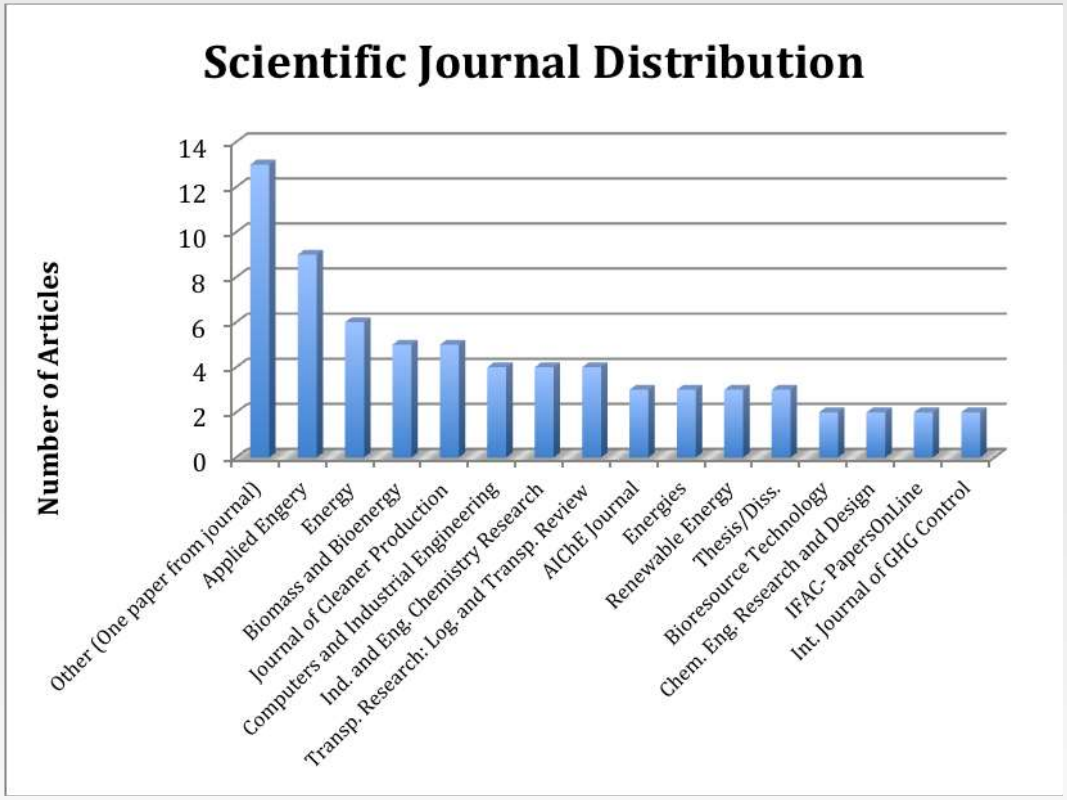


Figure . Scientific Journal Distribution

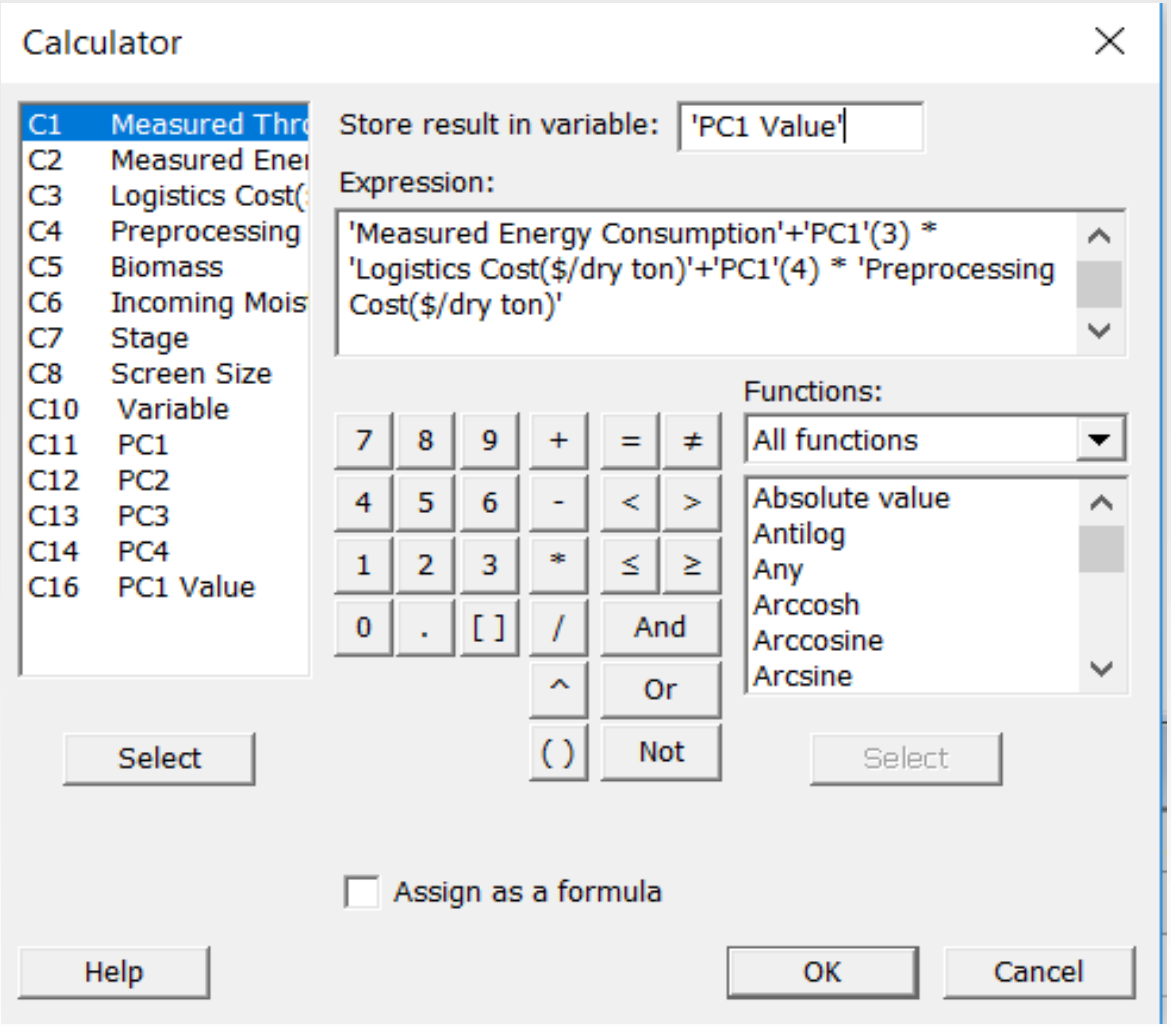
METHODOLOGY R Code

- **MOISTURE/BIOMASS NOT IN EXCEL**
- 1st download rgui, then r studio, r studio is what I used
- Then I saved all the files to the desktop, 3 different files, one with just the 4 components, another with the 4 components plus Biomass, then the third file was the 4 components plus moisture.
- To connect to excel go to session, set working directory, choose directory, then desktop.
- NEED TO INSTALL PACKAGES FOR PLOTS FIRST:
- Install.packages (“ggfortify”)
- Install.packages (“ggplot2”)
- Library(ggfortify)
- Library(ggplot2)
- NOW to start
- my.open = read.csv(“open.csv”) what is in “” is the name of excel file
- open.data <- read.csv (“open.csv”)
- open. Data
- my.prc = prcomp (open.data, center=T, scale=T) normalizes the data
- summary(my.prc) will give the stdv
- my.prc$sdev^2 will give the PC
- load = my.prc$rotation calculates eigenvectors
- load shows eigenvectors
- biplot(my.prc,cex=c(1,0.7)) bi plot but the minitab one looks better
- plot(my.prc) plots pcs vs variance

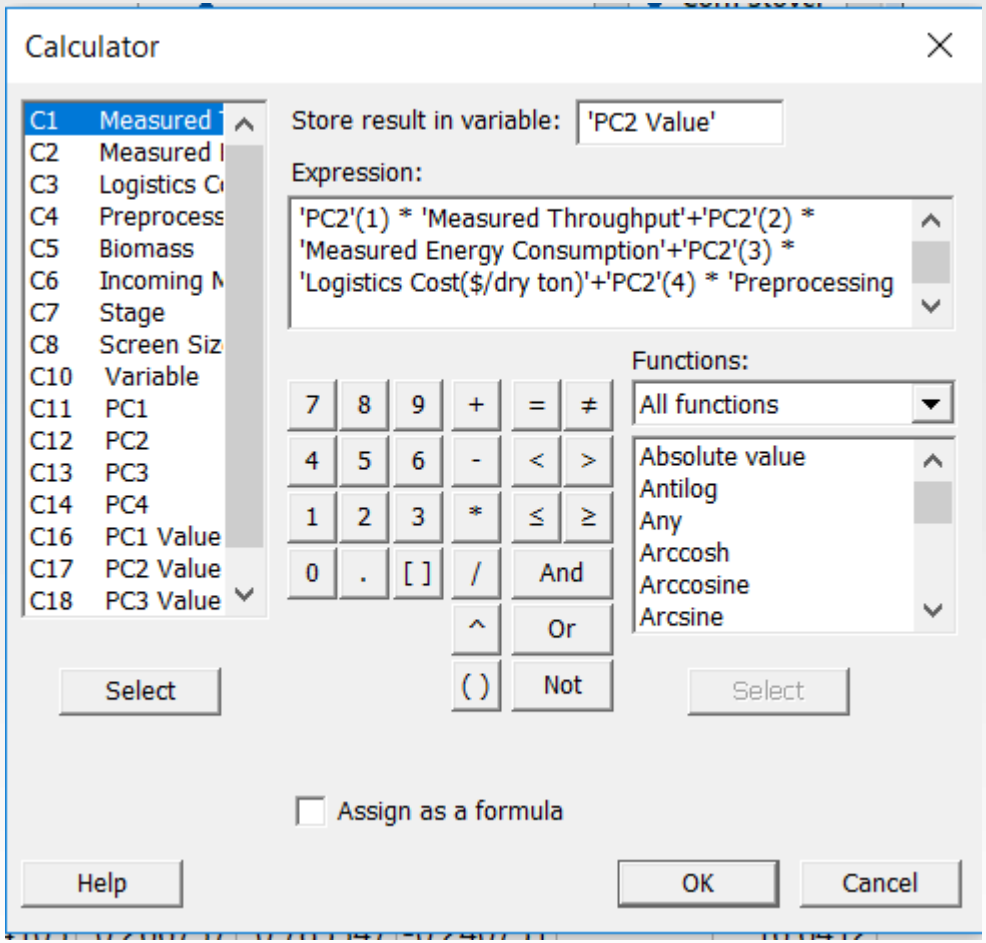
METHODOLOGY R Code

- **Autoplots:**
- **Have to upload the same way as above, first three lines, with either moisture or biomass. Each one was done separately.**
-
- My.open = read.csv (“Type.csv”)
- My.biomass <-read.csv(“Type.csv”)
- My.biomass shows the data to see if correct
- Biomass.data <-my.biomass[,1:4] separates the variables and factors
- Closed.biomass <-my.biomass[,5]
- My.pcabiomass <-prcomp (biomass.data, center = T, scale = T) gives the pca and normalizes the data
- autoplot(my.pcabiomass, data = my.biomass, colour = 'Biomass', loadings = TRUE)
- autoplot(my.pcabiomass, data = my.biomass, colour = 'Biomass', loadings = TRUE, loadings.colour = 'blue', load ings.label = TRUE, loadings.label.size = 3) gives the plots with labels.The one above did not give labels for pca
-
- Then do the same for the next set of data.

METHODOLOGY Minitab Steps

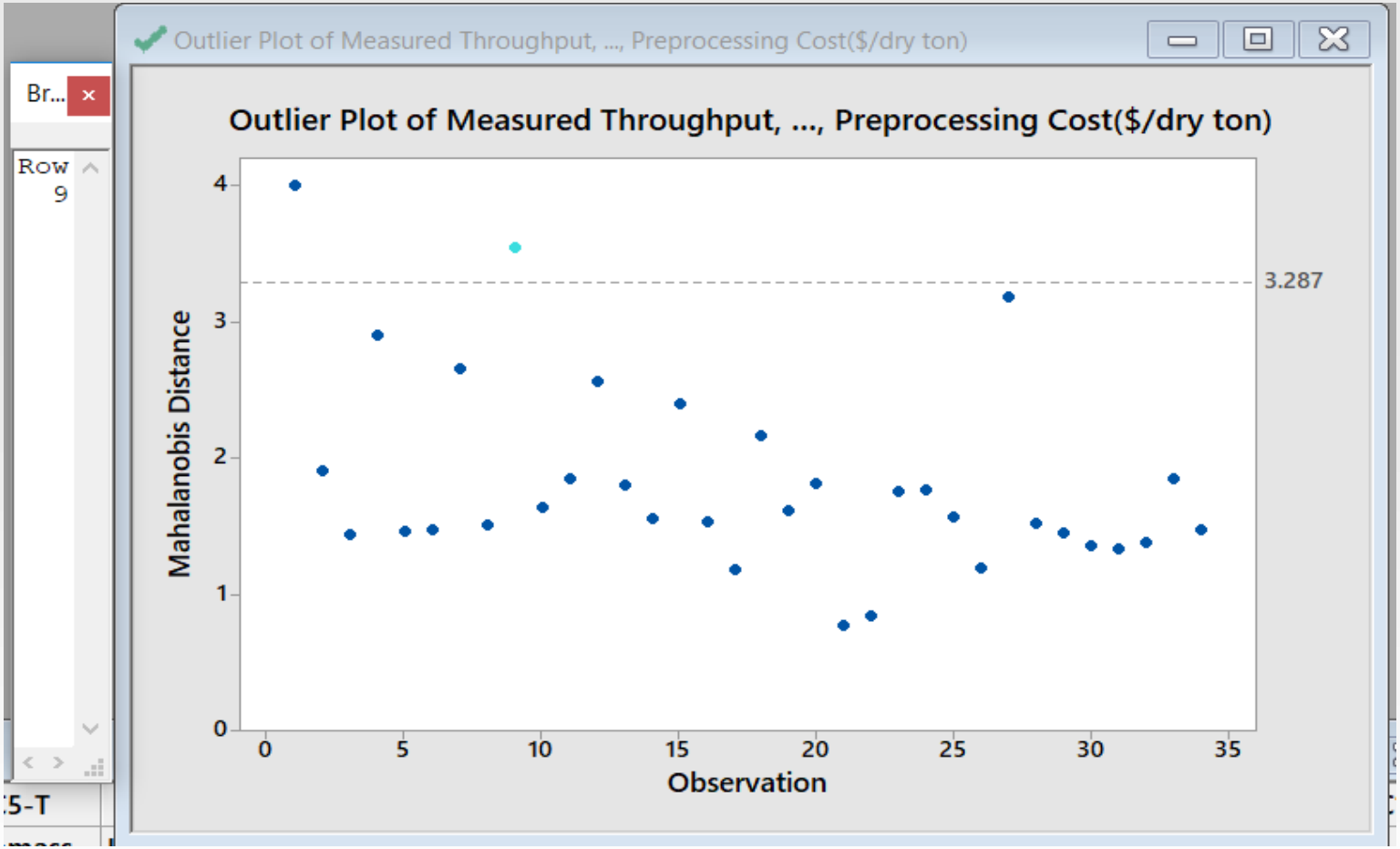


METHODOLOGY Minitab Steps

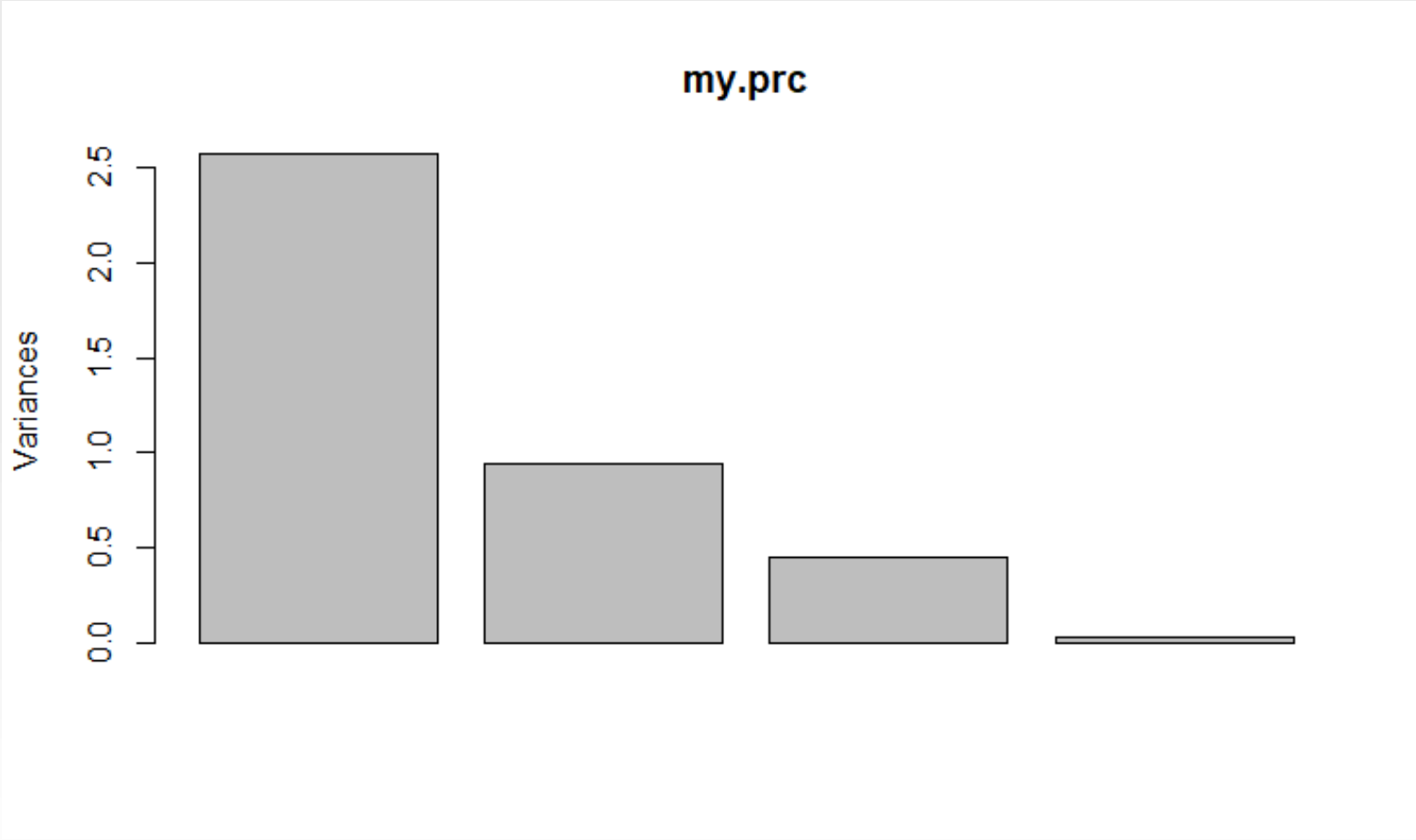


PC1 Value	PC2 Value	PC3 Value	PC4 Value
35.77562	-82.9755	32.68676	6.904962
16.6412	-41.4738	14.38856	9.081122
17.95809	-44.6039	14.85056	9.296044
25.80434	-40.4883	24.70395	7.059066
18.0327	-44.8608	14.89497	9.312534
16.64807	-39.2062	13.84802	9.263048
30.98991	-61.7705	28.65127	7.214057
19.52542	-52.2284	16.34283	8.998291
29.68586	-86.3748	22.25368	9.604204
16.61043	-41.5472	14.03506	9.028255
18.20578	-44.5545	14.79994	9.081228
19.0857	-44.008	14.42606	10.30961
18.28039	-44.8114	14.84435	9.097718
16.61035	-39.2561	13.62919	9.245674
24.90015	-65.1697	18.21819	9.913299
19.4877	-52.2783	16.124	8.980916
24.151	-42.4442	18.81227	9.708361
22.16497	-32.8105	19.62122	8.311888
22.42132	-34.9687	17.40108	9.765557
22.07737	-33.826	18.32201	8.995763
25.49393	-44.5949	20.78152	9.250771
24.97189	-42.4218	20.37644	9.244037
26.89439	-52.6657	20.44988	10.0514
27.65551	-47.8523	21.25277	9.877547
26.62333	-51.8221	20.31104	9.965557
26.43303	-41.1714	22.71668	8.695717
20.5736	-33.4811	18.05866	9.274599
24.70335	-33.6959	21.30549	8.752913
21.62891	-33.9653	18.14715	9.325767
27.06222	-43.7079	23.53033	8.569347
26.54018	-41.5348	23.12525	8.562612
29.06551	-51.5305	23.75953	8.998425
27.55643	-47.872	21.27302	9.963473

METHODOLOGY Minitab Steps



METHODOLOGY Minitab Steps



CASE STUDY Data Collection

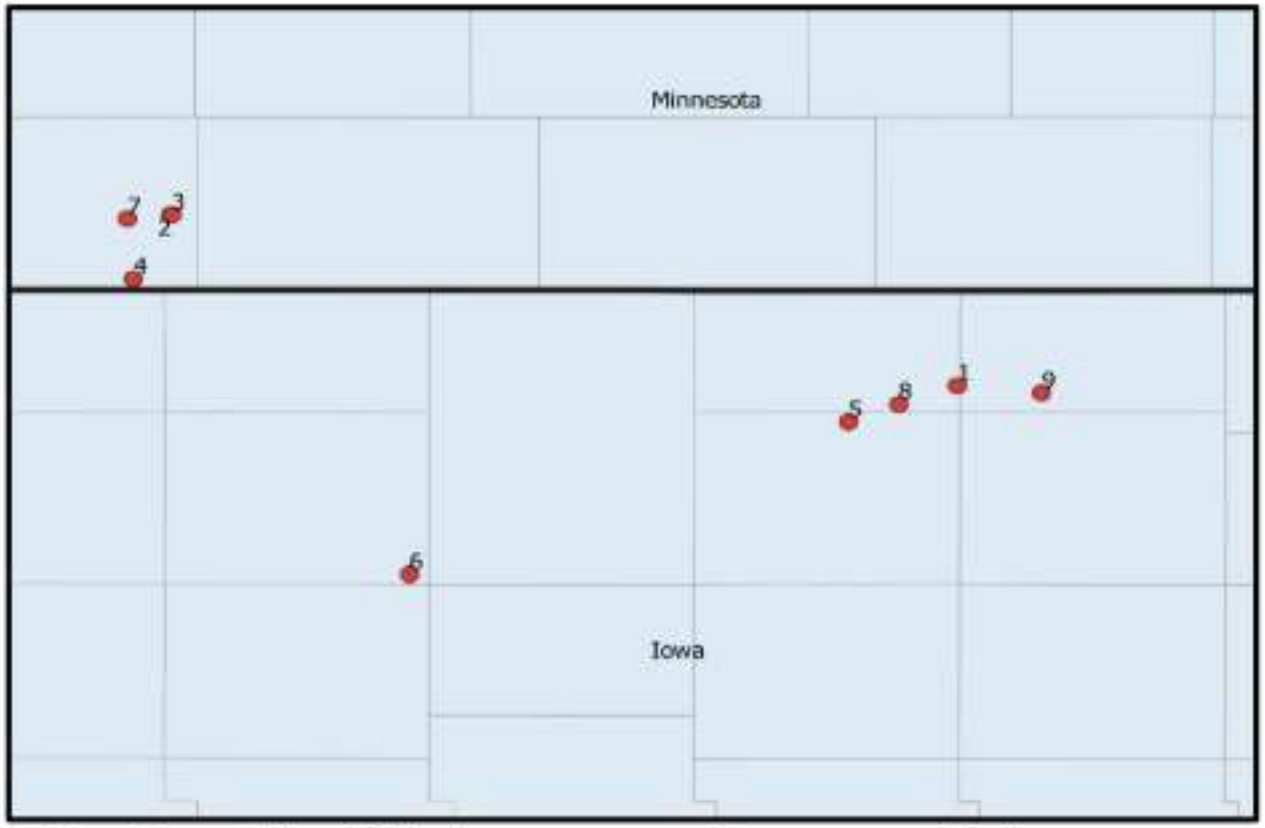


Figure 2: Selected locations of nine fields for corn-stover base case analysis.

CASE STUDY Data Collection

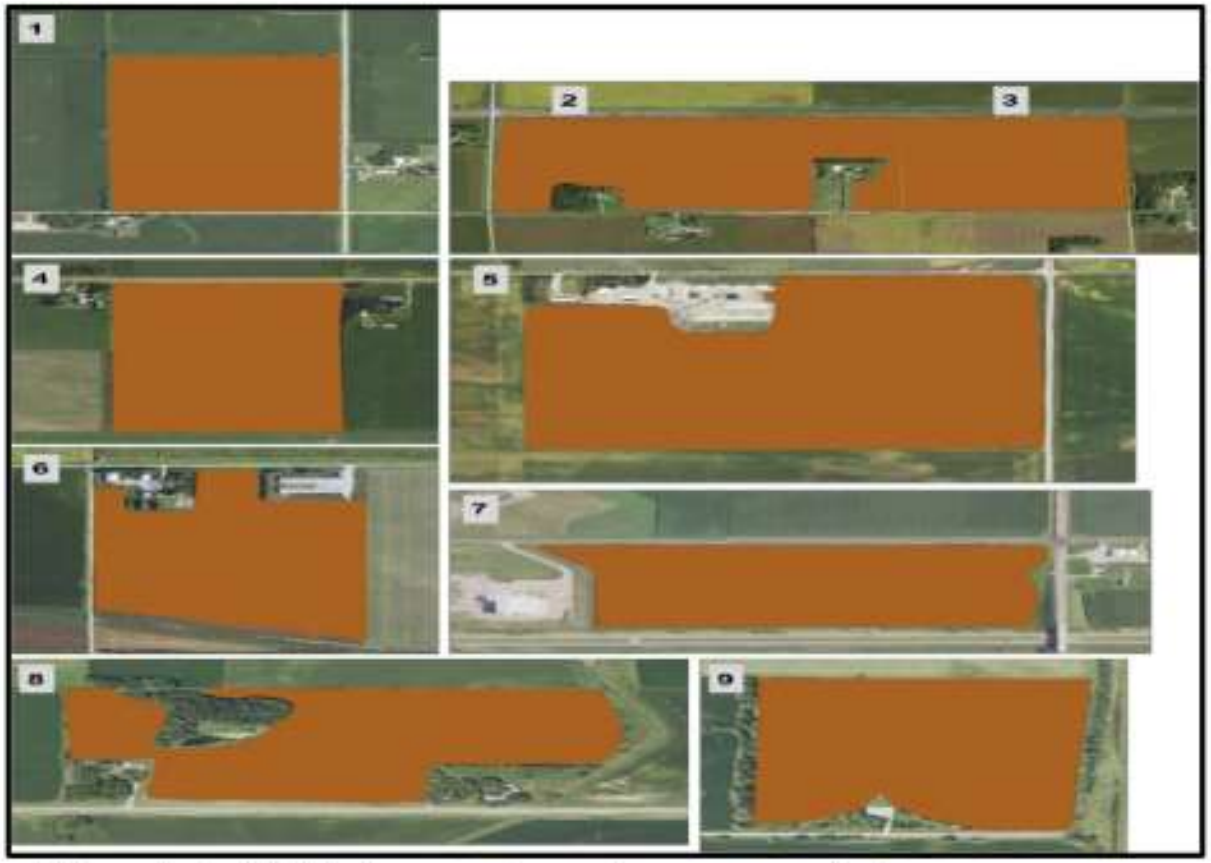


Figure 3: Shape of the selected fields for corn stover base case analysis.

CASE STUDY Data Collection

Table C-2. Cost breakdown by operations for different fields of corn stover.

Field name	Field 1	Field 2	Field 3	Field 4	Field 5	Field 6	Field 7	Field 8	Field 9
Harvest and collection cost($/dry ton)	**$19.21**	**$12.45**	**$12.92**	**$12.13**	**$21.58**	**$19.37**	**$26.61**	**$23.28**	**$28.08**
Baling	$15.84	$9.01	$9.74	$8.76	$18.15	$15.88	$22.74	$19.78	$24.84
Roadsider	$3.37	$3.44	$3.18	$3.37	$3.43	$3.49	$3.87	$3.50	$3.24
Field storage	**$3.60**	**$3.05**	**$2.94**	**$2.98**	**$3.85**	**$3.68**	**$4.52**	**$4.08**	
Stack	$3.60	$3.05	$2.94	$2.98	$3.85	$3.68	$4.52	$4.08	$4.30
Transportation/handling cost from field to biorefinery	$8.51	$7.97	$7.97	$8.49	$8.61	$8.79	$9.74	$8.84	$8.13
Trucking/unloading	$8.51	$7.97	$7.97	$8.49	$8.61	$8.79	$9.74	$8.84	$8.13
Preprocessing at biorefinery	**$15.42**	**$13.74**	**$15.25**	**$14.37**	**$15.36**	**$15.28**	**$16.26**	**$16.82**	**$16.12**
Grinder 1	$8.74	$8.12	$8.74	$8.41	$8.64	$8.64	$9.00	$9.75	$9.00
Grinder 2	$4.56	$3.65	$4.56	$3.99	$4.51	$4.51	$4.87	$4.78	$4.87
Conveyors	$0.26	$0.26	$0.24	$0.26	$0.26	$0.27	$0.38	$0.33	$0.24
Dust collection	$1.59	$1.44	$1.44	$1.44	$1.68	$1.59	$1.74	$1.69	$1.74
Surge bin	$0.05	$0.05	$0.05	$0.05	$0.05	$0.05	$0.05	$0.05	$0.05
Misc. Equipment[a]	$0.22	$0.22	$0.22	$0.22	$0.22	$0.22	$0.22	$0.22	$0.22
Reactor in-feed and storage	**$2.29**	**$2.34**	**$2.15**	**$2.28**	**$2.32**	**$2.36**	**$2.59**	**$2.38**	**$2.18**
Storage at refinery	$0.94	$0.96	$0.88	$0.93	$0.95	$0.97	$1.03	$0.98	$0.89
Handling and queuing	$1.35	$1.38	$1.27	$1.35	$1.37	$1.39	$1.56	$1.40	$1.29
Logistics cost	**$49.03**	**$39.55**	**$41.23**	**$40.25**	**$51.72**	**$49.48**	**$59.71**	**$55.40**	**$54.52**

a: Miscellaneous equipment consists of moisture meters, bale rejecters, electromagnets, etc.

CASE STUDY Data Collection

Cost Breakdown by Operation

Table C-1. Cost breakdown by operations for different fields of switchgrass.

Field name	Merck 4	Merck 6	Merck 7	Merck 8	Merck 9	Merck 10	Merck 13	Merck 14
Harvest and collection	$60.33	$19.35	$21.75	$19.60	$22.03	$17.33	$40.69	$32.34
Mowing	$10.89	$2.27	$3.25	$2.76	$3.99	$3.49	$14.10	$3.79
Conditioning with mower	$13.61	$2.67	$3.88	$3.63	$1.55	$0.00	$6.07	$1.59
Raking	$6.95	$1.36	$1.98	$1.85	$2.45	$2.40	$9.61	$2.52
Baling	$24.99	$9.34	$8.87	$7.52	$10.33	$7.93	$7.39	$21.53
Roadsider	$3.89	$3.71	$3.77	$3.84	$3.71	$3.51	$3.52	$2.91
Field storage	$5.69	$3.34	$3.47	$3.37	$3.62	$3.28	$4.59	$3.92
Stack	$5.69	$3.34	$3.47	$3.37	$3.62	$3.28	$4.59	$3.92
Transportation and handling from field to birefinery	$9.74	$9.30	$9.49	$9.66	$9.30	$8.81	$8.86	$7.12
Trucking/unloading	$9.74	$9.30	$9.49	$9.66	$9.30	$8.81	$8.86	$7.12
Preprocessing at biorefinery	$14.66	$11.24	$11.88	$13.47	$11.91	$11.66	$14.37	$11.74
Grinder 1	$8.03	$6.13	$6.45	$8.03	$6.45	$6.45	$8.03	$6.45
Grinder 2	$3.84	$3.02	$3.27	$3.27	$3.27	$3.13	$3.84	$3.13
Conveyors	$0.38	$0.33	$0.35	$0.36	$0.33	$0.29	$0.31	$0.19
Dust collection	$2.14	$1.49	$1.54	$1.54	$1.59	$1.52	$1.92	$1.70
Surge bin	$0.05	$0.05	$0.05	$0.05	$0.05	$0.05	$0.05	$0.05
Misc. Equipments[a]	$0.22	$0.22	$0.22	$0.22	$0.22	$0.22	$0.22	$0.22
Reactor in-feed and storage	$2.58	$2.50	$2.50	$2.55	$2.50	$2.37	$2.38	$1.90
Storage at refinery	$1.02	$1.03	$0.99	$1.01	$1.03	$0.97	$0.98	$0.77
Handling and queuing at refinery	$1.56	$1.47	$1.51	$1.54	$1.47	$1.40	$1.40	$1.13
Logistics cost	$93.00	$45.73	$49.09	$48.65	$49.36	$43.45	$70.89	$57.02

a: Miscellaneous equipment consists of moisture meters, bale rejecters, electromagnets, etc.

CASE STUDY Data Collection

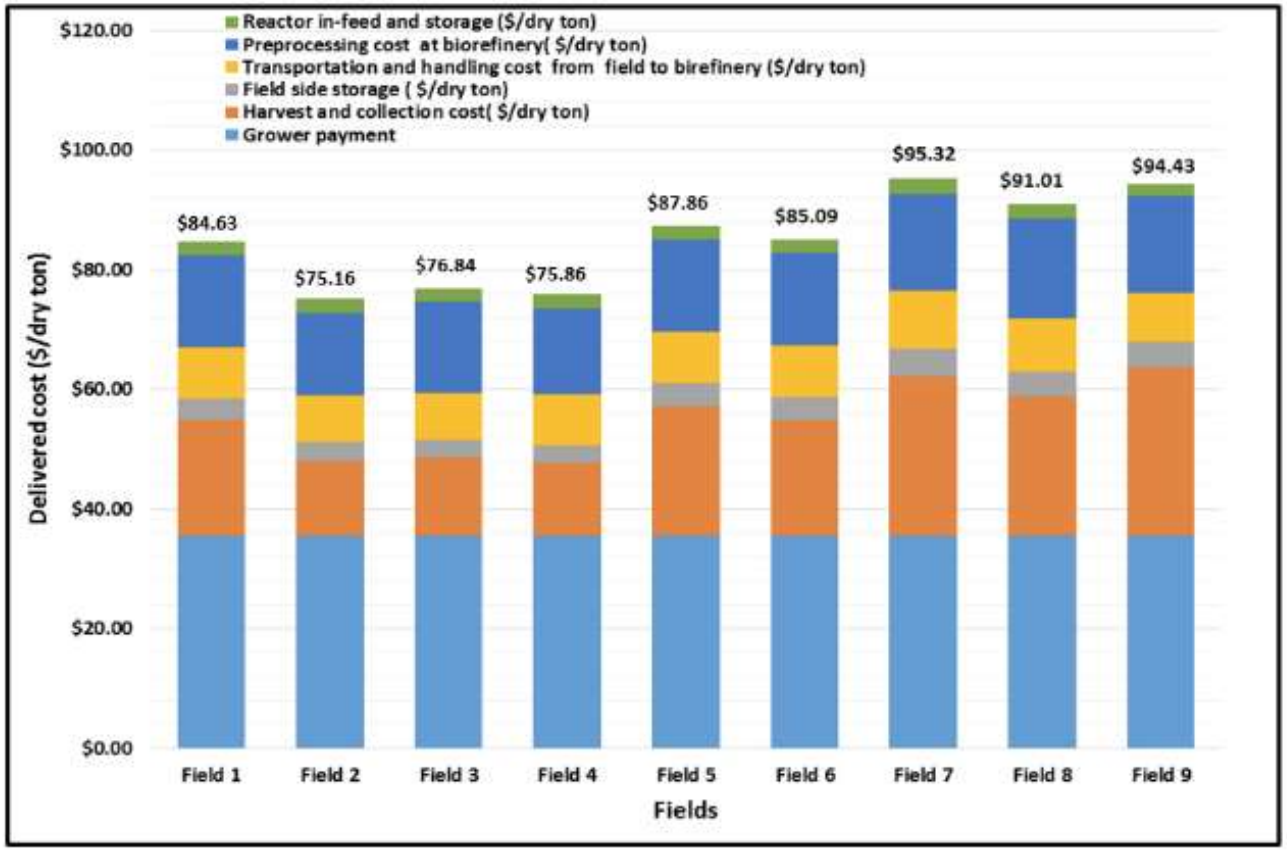


Figure 9 : Delivered feedstock cost from different fields assuming a grower payment of $35.61/dry ton.